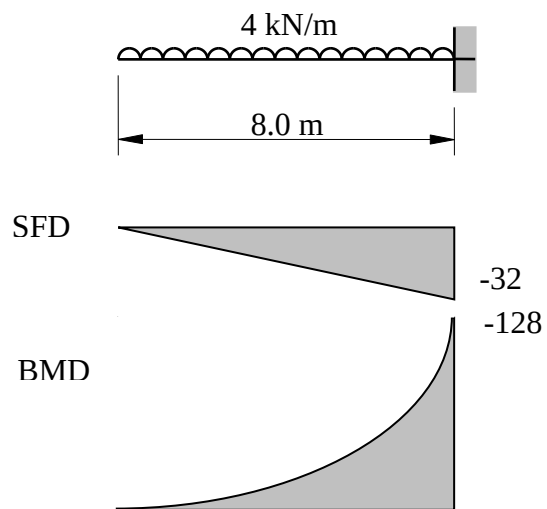


THE UNIVERSITY OF BIRMINGHAM
DEPARTMENT OF CIVIL ENGINEERING

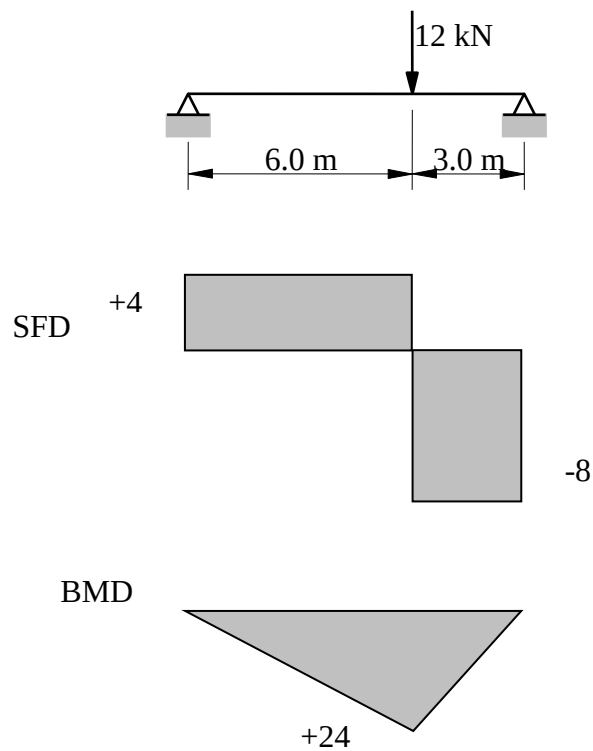
Mechanics 1

Tutorial Sheet - 4

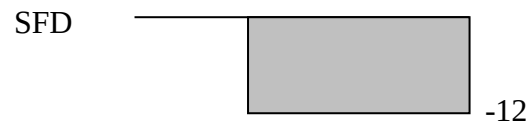
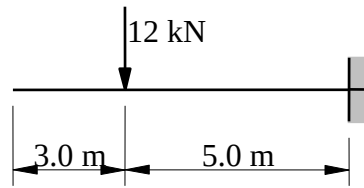
1a



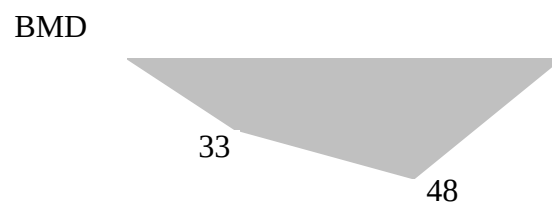
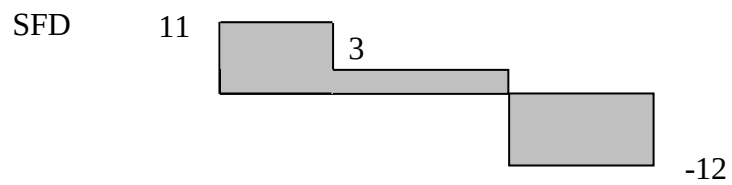
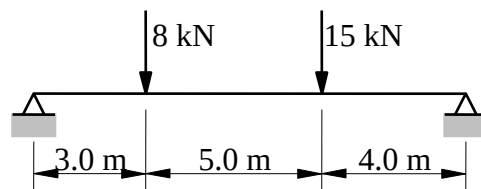
1b



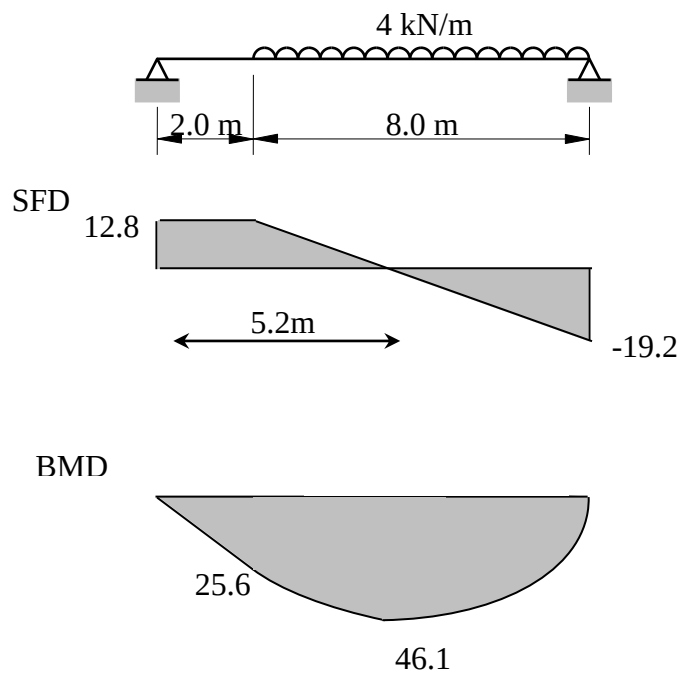
1c



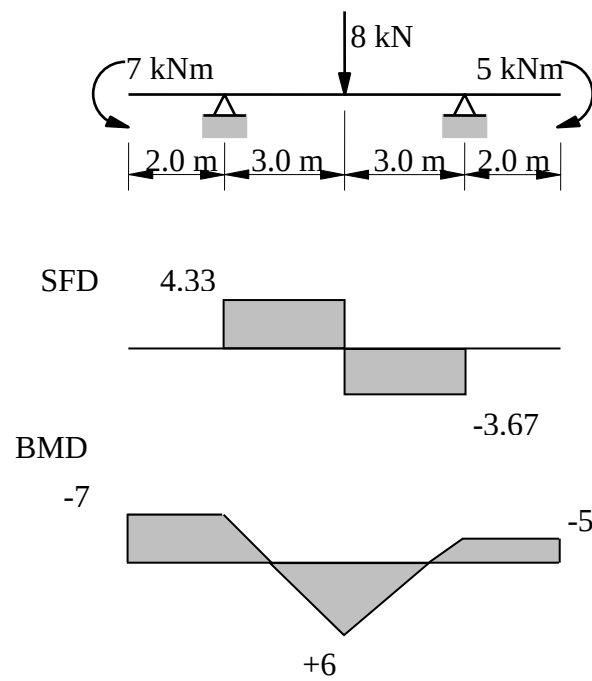
1d



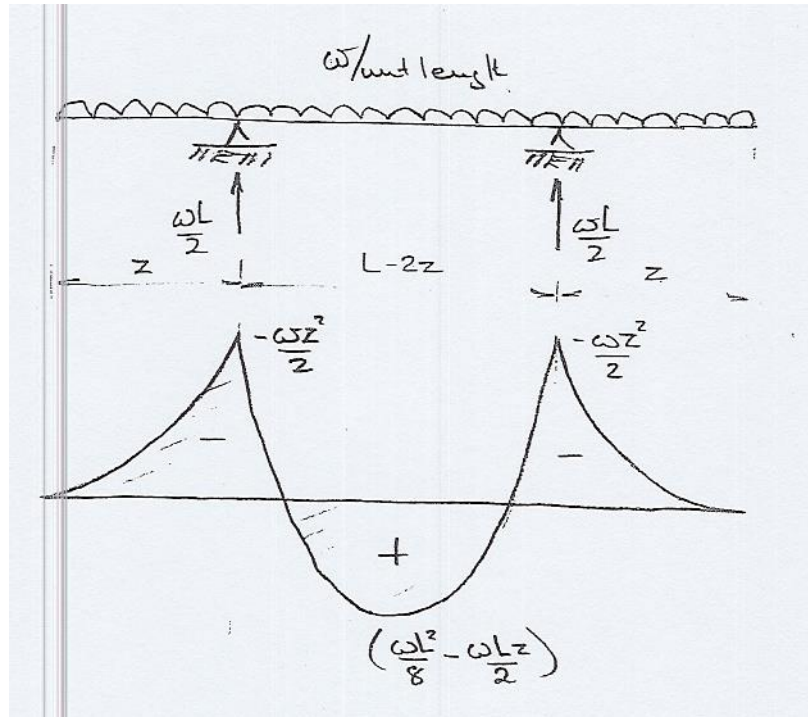
1e



1f



2



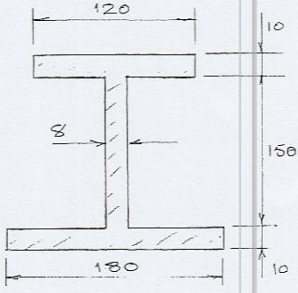
As z increases – the sagging moment becomes smaller; the hogging moment becomes larger. The larger of the two is as small as possible when they are equal:

$$\frac{wL^2}{8} - \frac{wLz}{2} = \frac{wz^2}{2}$$

$$4z^2 + 4Lz - L^2 = 0 \quad \Rightarrow \quad \frac{z}{L} = \frac{-4 \pm \sqrt{16+16}}{8} = \frac{\sqrt{2}-1}{2} = 0.207$$

3a

(a)



$$A = 1200 + 1200 + 1800 = 4200 \text{ mm}^2$$

$$\bar{y} = \frac{(1200 \cdot 5 + 1200 \cdot 85 + 1800 \cdot 165)}{4200} = \underline{\underline{96.429 \text{ mm}}}$$

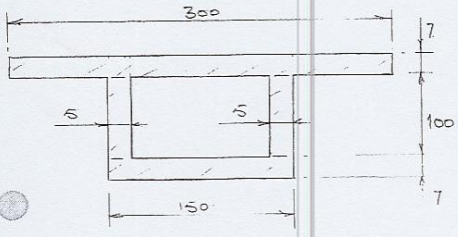
$$I = \left(\frac{1}{12} \cdot 120 \cdot 10^3 + 1200 \cdot 91.429^2 \right) + \left(\frac{1}{12} \cdot 8 \cdot 150^3 + 1200 \cdot 11.429^2 \right) + \left(\frac{1}{12} \cdot 180 \cdot 10^3 + 1800 \cdot 68.571^2 \right)$$

$$= \underline{\underline{2.093 \cdot 10^7 \text{ mm}^4}}$$

$$\sigma = \frac{10 \cdot 10^6 \cdot -96.429}{2.093 \cdot 10^7} = \underline{\underline{-46.07 \text{ N/mm}^2}} ; \quad = \frac{10 \cdot 10^6 \cdot 73.571}{2.093 \cdot 10^7} = \underline{\underline{+35.15 \text{ N/mm}^2}}$$

3b

b)



$$A = 2100 + 1000 + 1050 = 4150 \text{ mm}^2$$

$$\bar{y} = \frac{(2100 \cdot 3.5 + 1000 \cdot 57 + 1050 \cdot 110.5)}{4150} = \underline{\underline{43.464 \text{ mm}}}$$

$$I = \left(\frac{1}{12} \cdot 300 \cdot 7^3 + 2100 \cdot 39.964^2 \right) + \left(\frac{1}{12} \cdot 10 \cdot 100^3 + 1000 \cdot 13.536^2 \right) + \left(\frac{1}{12} \cdot 150 \cdot 7^3 + 1050 \cdot 67.036^2 \right)$$

$$= \underline{\underline{9.102 \cdot 10^6 \text{ mm}^4}}$$

$$\sigma = \frac{10 \cdot 10^6 \cdot -43.464}{9.102 \cdot 10^6} = \underline{\underline{-47.75 \text{ N/mm}^2}} ; \quad = \frac{10 \cdot 10^6 \cdot 70.536}{9.102 \cdot 10^6} = \underline{\underline{+77.50 \text{ N/mm}^2}}$$

4

Positive B.Mt — tension on the bottom :-

$$(a) \quad M = \frac{+100 \cdot 20.93 \cdot 10^6}{+73.571} = +28.3 \text{ kNm} ; \quad M = \frac{-70 \cdot 20.93 \cdot 10^6}{-96.429} = +15.2 \text{ kNm}$$

$$(b) \quad M = \frac{+100 \cdot 9.102 \cdot 10^6}{+70.536} = +12.9 \text{ kNm} ; \quad M = \frac{-70 \cdot 9.102 \cdot 10^6}{-43.464} = +14.7 \text{ kNm}$$

Negative B.Mt — tension at the top :-

$$(a) \quad M = \frac{+100 \cdot 20.93 \cdot 10^6}{-96.429} = -21.7 \text{ kNm} ; \quad M = \frac{-70 \cdot 20.93 \cdot 10^6}{+73.571} = -19.9 \text{ kNm}$$

$$(b) \quad M = \frac{+100 \cdot 9.102 \cdot 10^6}{-43.464} = -20.9 \text{ kNm} ; \quad M = \frac{-70 \cdot 9.102 \cdot 10^6}{+70.536} = -9.0 \text{ kNm}$$

5

$$\begin{aligned} \text{C.S.A of pipe} &= 7740.9 \text{ mm}^2 = \frac{\pi}{4} (316^2 - 300^2) \\ \text{of fluid} &= 70685.9 \text{ mm}^2 = \frac{\pi}{4} \cdot 300^2 \end{aligned}$$

$$\begin{aligned} \therefore \text{Mass} &= (7.8 \times 7740.9 + 1.040 \cdot 70685.9) \times 10^{-6} \times 1000 \\ &= 133.832 \text{ kg/m} \end{aligned}$$

$$\text{Weight} = 133.832 \times 9.81 = 1313.48 \text{ N/m}$$

Simply-supported, span 12.0 m :-

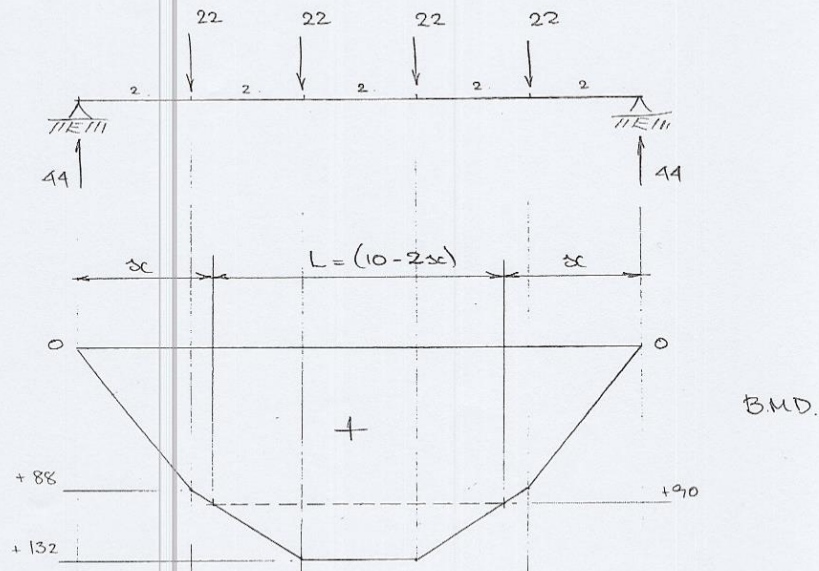
$$\begin{aligned} \text{Max. B.Mt} &= \frac{wl^2}{8} = \frac{1313.48 \cdot 12^2}{8} \\ &= 23642.64 \text{ N.m} \end{aligned}$$

$$I = \frac{\pi}{64} (316^4 - 300^4) \times 10^{-12} = 91.853 \times 10^{-6} \text{ m}^4$$

$$\begin{aligned} \text{Stress, } \sigma &= \frac{My}{I} = \frac{23642.64 \times 0.316/2}{91.853 \times 10^{-6}} \\ &= 40.67 \times 10^6 \text{ N/m}^2 \\ &= \underline{\underline{40.67 \text{ N/mm}^2}} \end{aligned}$$

6

Max moment $M = \frac{\sigma I}{y} = \frac{100 \cdot 1.80 \times 10^8}{200} = 0.90 \cdot 10^8 \text{ N.mm}$
 $= \underline{\underline{90 \text{ kN.m.}}}$



Max. mt. on reinforced beam = 132 kN.m.

$\therefore$ 2nd Mt reqd. $I = \frac{M_{\max} \cdot y}{\sigma} = \frac{132 \cdot 10^6 \cdot 210}{100} = 2.772 \cdot 10^8 \text{ mm}^4$

$I = \{1.80 \cdot 10^8 + 2(b \cdot 10 \cdot 205^2)\} = 2.772 \cdot 10^8$

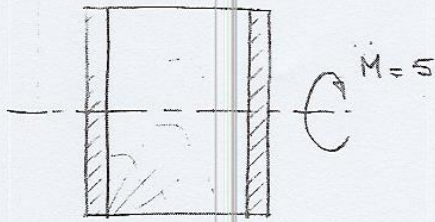
$b = \frac{(2.772 - 1.80) \cdot 10^8}{2 \cdot 10 \cdot 205^2} = \underline{\underline{115.6 \text{ mm}}}$

B.Mt of +90 kN.m occurs at :-

$+44x - 22(x-2) = +90 \quad ; \quad x = 2.091 \text{ m.}$

$\therefore L = 10 - 2x = 10 - 2 \cdot 2.091 \cdot 2$

$= \underline{\underline{5.818 \text{ m.}}}$



M_s = moment carried by steel

M_t = _____ timber

Equilibrium : $M_t + M_s = 5$

Elasticity : $\frac{1}{R_t} = \frac{M_t}{E_t I_t}$; $\frac{1}{R_s} = \frac{M_s}{E_s I_s}$

Compatibility : $\frac{1}{R_t} = \frac{1}{R_s}$

$$\frac{M_t}{M_s} = \frac{E_t I_t}{E_s I_s} = \frac{12}{200} \cdot \frac{\frac{1}{12} 50 \cdot 100^3}{\frac{1}{12} 14 \cdot 100^3} = 0.214$$

$$1.214 M_s = 5 \quad ; \quad M_s = \underline{\underline{4.118 \text{ kNm}}}$$

$$M_t = \underline{\underline{0.882 \text{ kNm}}}$$

$$\sigma_s = \frac{M_s \cdot 50}{I_s} = \frac{4.118 \cdot 10^6 \cdot 50}{\frac{1}{12} 14 \cdot 100^3} = \underline{\underline{176.5 \text{ N/mm}^2}}$$

$$\sigma_t = \frac{M_t \cdot 50}{I_t} = \frac{0.882 \cdot 10^6 \cdot 50}{\frac{1}{12} 50 \cdot 100^3} = \underline{\underline{10.6 \text{ N/mm}^2}}$$